

Computer Vision Assignment

CYSHIELD

Task code	CVCY002	Position:	Computer Vision Engineer
Objective: The goal of this assessment is to develop a clothes segmentation model that accurately segments the clothes from people.			
Introduction:	This assessment challenges to develop a robust clothing segmentation model capable of accurately separating clothes from images of people . clothing segmentation has become a crucial component for virtual fitting rooms. The task involves selecting an appropriate dataset, implementing a segmentation model and driving a pipeline to segment worn clothes from people's images.		
Assignment Tasks:			
1. Dataset Selection & Preprocessing			
a. Choose a proper dataset. b. Preprocess and handle data augmentation			
2. Model Development			
a. Train deep learning-based model for clothes segmentation. b. Drive pipeline to take personal image and segment clothes from the image.			
3. Discover System Capabilities			
a. List the strength of the system, discover how powerful the model to detecting and segmenting the clothes. b. List drawbacks of the system, discover the weakness of the model and wrong segmentation. c. Set limitations for your system (e.g., conditions of capturing.)			
4. Deliverables:			
You must submit the following in a GitHub repository: A well-documented script containing: a. Dataset preprocessing b. Model implementation c. Training and evaluation code A report (PDF or Markdown) explaining: a. Dataset choice and reason b. Model architecture c. Loss function selection and reason d. Performance analysis and evaluation metrics. e. Limitations of the system A README.md with clear instructions on how to set up, run the code, and reproduce results.			
Note!	The system must achieve at least the minimum required performance for the target task		

